

standard form

$$\Phi(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

damping classes

$\zeta = 0$: undamped

$0 < \zeta < 1$: underdamped

$\zeta = 1$: critical damping

$\zeta > 1$: overdamped